Midterm EDA Report - OMDB Exploration

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Table of Contents

1. Introduction **2**

1.1 Inspiration

1.2 Guiding Questions

2. Methodology **3**

3. Sourcing **3**

3.1 OMDB API

3.2 Input list - movies.csv

4. Variables **4**

4.1 OMDB API Built-In Variables

4.2 Custom Variables

5. Data Cleaning **7**

6. Exhibits and Interpretation **7**

7. Discussion **18**

8. Conclusion **19**

1. Introduction

1.1 Inspiration

Our inspiration for exploring this dataset stemmed from a mutual love for movies, not just surface-level aspects of them, though, rather their ability to put us on a roller coaster of emotions and feelings, and we wanted to see if there was a way to quantify this. We also use the app 'Letterboxd', which is a social media for rating movies with your friends, and sharing reviews on them. This was in part the reason we focused on ratings and different directors/actors, as we had been friends on the app and knew this was a shared interest.

1.2 Guiding Questions

Which numeric variables are most positively or negatively correlated with box office performance in the US?

How do movies perform at the box office across different primary actors?

How does the box office performance of different notable directors differ?

What is people's average opinion of films from each decade? Is there a trend in how "good" films are over time?

How does whether or not a film is offered in English affect its box office, average rating, and total votes performance?

How does whether or not a film is appropriate for children affect its box office, rating, and popularity performance?

Is there any correlation between how different critics rate a film and how it performed at the box office?

Have films typically gotten longer or shorter over time?

Is there a relationship between a film's IMDb rating and its overall popularity, as measured by total votes?

How closely do critic ratings (Metascore) and audience ratings (IMDb) align, and which films show the greatest divergence between the two?

How does a film being offered in English affect its box office performance, regardless of its IMDb rating?

2. Methodology

To explore this data set, we used the OMDB API, a large movie database with all the information we were looking for (IMDb ratings, genres, Rotten Tomatoes scores, etc.). With their built-in features, their own categorical and calculated columns, and their storage in a dataframe, we could examine their capabilities with ease and in an organized manner. In order to visualize the data, we used various libraries, including pandas, seaborn, numpy, datetime, and matplotlib.

3. Sourcing

3.1 OMDB API

In order to actually access data on our input films, we used the free OMDB API, which provides information on US film performance, reviews, and a variety of basic information.

Documentation for the API can be found here: https://www.omdbapi.com/

3.2 Primary Input - *movies.csv*

Our main source of actual data consisted of a list of 338 films that we have either watched or want to watch. These films ranged from domestic blockbuster hits to foreign arthouse cult classics

and from the 1920s to modern times. We then made this list into a CSV and made sure the data was as clean as possible for use with the API by removing any non-Latin characters and trailing spaces.

4. Variables

4.1 OMDB API Built-In Variables

1. ***Movie***
    a. Title of the movie, stored as a string.
2. ***Year***
    a. Year of production, stored as a year datetime value.
3. ***Released***
    a. The exact release date of the film in the US domestic market is stored as a datetime value.
4. ***Runtime (min)***
    a. Length of the film in minutes, stored as a float
5. ***Rating***
    a. Age-rating of the movie, stored as a string.
6. ***Votes***
    a. Number of votes received on IMDB (relative measure of popularity), stored as a float. This acts as a relative measure of popularity.
7. ***Director***
    a. Name of the director(s) of the film, stored as a string.
8. ***Actors***
    a. Names of the leading actors, stored as a string with each name separated by a comma.

9. ***Country***

    a. Country or countries of origin for the film, stored as a string.

10. ***Language***

    a. Languages that the film was originally offered in, stored as a string.

11. ***IMDB Score***

    a. IMDB average rating out of 10, stored as a float.

12. ***Metascore***

    a. Average Metascore out of 100, stored as a float.

13. ***Box Office (usd)***

    a. US box office performance of the film in dollars, stored as a float.

14. ***Awards***

    a. String value describing the number of awards received by the film and from where.

4.2 Custom Variables

15. ***Decade***

    a. Decade in which the film was produced, stored as a string.

16. ***Categorical Length***

    a. Short

        i. Categorical variable, string value. Movies were classified as short if their runtime was under 90 minutes.

    b. Medium

        i. Categorical variable, string value. Movies were classified as medium if their runtime was between 90 and 150 minutes in length.

    c. Long

i. Categorical variable, string value. Movies were classified as long if their runtime was longer than 150 minutes in length.

17. Primary Genre

a. String value that stores the first genre listed in the genre variable.

18. Secondary Genre

a. String value that stores the second genre listed in the genre variable, if applicable.

19. Tertiary Genre

a. String value that stores the third genre listed in the genre variable, if applicable.

20. Primary Language

a. String value that stores the first language listed in the language variable.

21. Secondary Language

a. String value that stores the second language listed in the language variable, if applicable.

22. is_English

a. True

i. Boolean variable, True if the movie is offered in English

b. False

i. Boolean variable, False if the movie is not offered in English

23. Primary Actor

a. String value that stores the first actor listed in the actors variable.

24. Secondary Actor

a. String value that stores the second actor listed in the actors variable.

25. Tertiary Actor

a. String value that stores the third actor listed in the actors variable.

26. is_Appropriate

a. True

i. Boolean variable, True if the movie is rated PG-13, PG, or G.

b. False

i. Boolean variable, False if otherwise, as the movie would likely not be appropriate for children.

27. Average Rating

a. Float value, the weighted average of the IMDB score and Metascore out of 100.

28. Difference in Rating

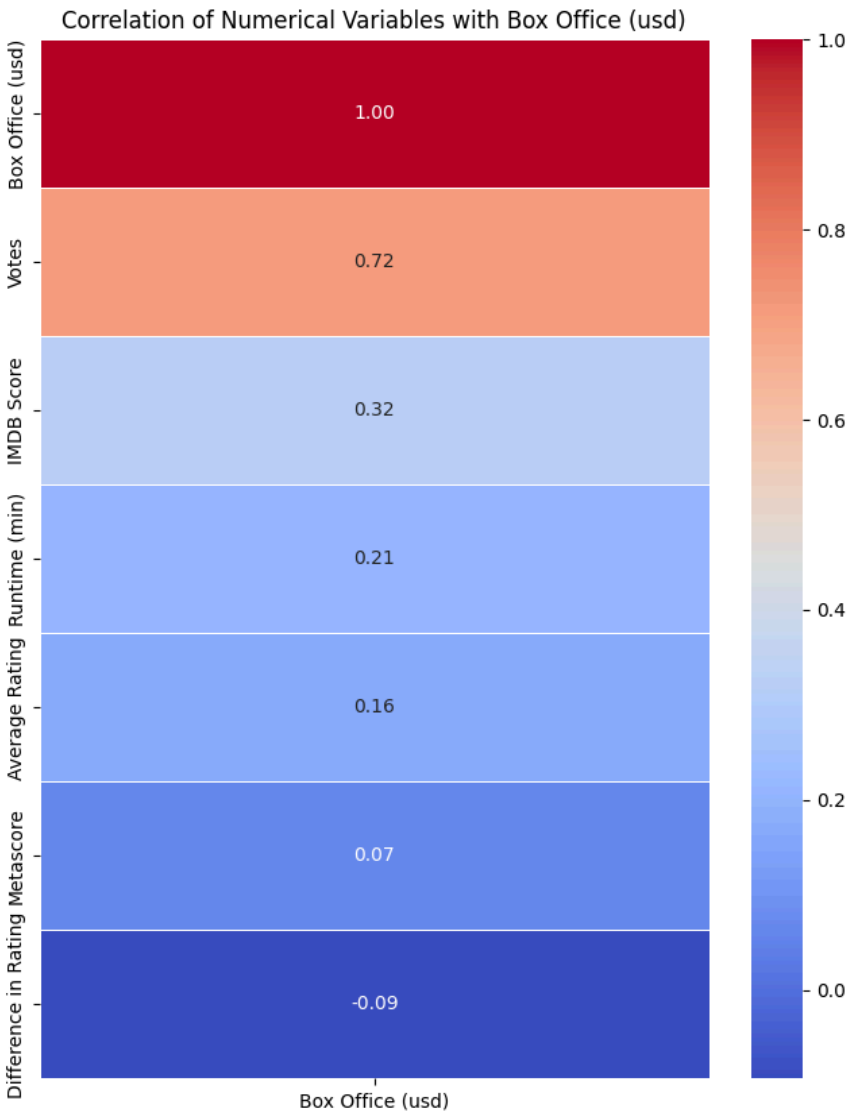
a. Float value, the absolute value of the difference between the Metascore and the IMDB score times ten.

5. Data Cleaning

In order to make logical sense of much of the data, we had to take a few steps to clean some of the different columns. For example, we converted release date and production year to datetime values to better track film performance over time. Additionally, any numeric value, such as box office performance or metascore, originally loaded in as a string, required a conversion to a float. Moreover, we created a few categorical variables to better make sense of columns like rating and runtime.

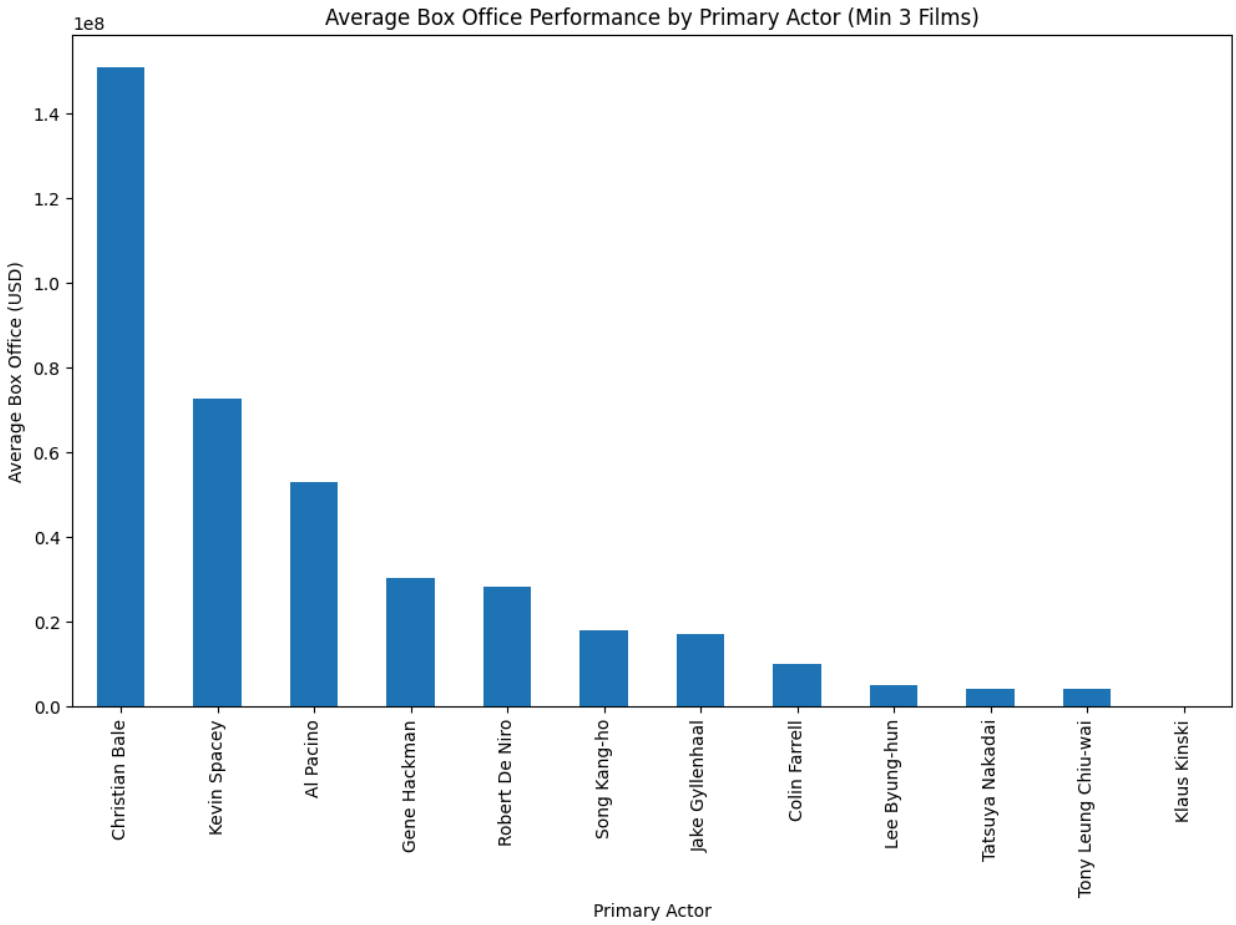
6. Exhibits and Interpretation

Fig. 6.1: *Correlation of Numerical Variables with US Box Office Performance*



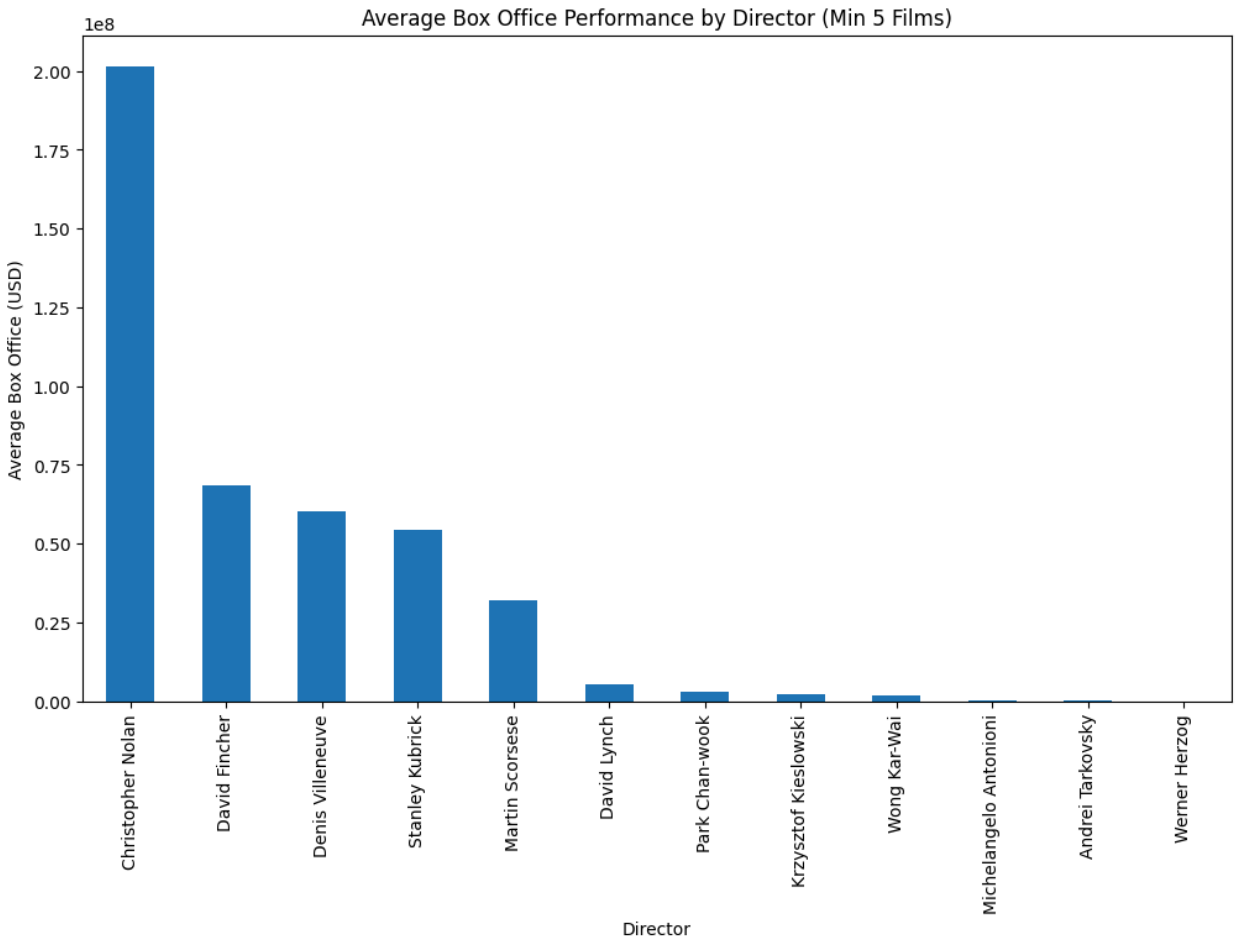
This figure analyzes the correlation between box office performance and the different numeric variables in the data. Looking at this chart, we see that "votes" has the most significant positive correlation with box office performance, which makes sense, as votes and box office performance are both relative measures of popularity. The second most correlated variable, though far off from votes, is IMDB score, which is somewhat logical that we would see a positive correlation here, as better movies should or would have fared better at theaters. The only variable with a slight negative correlation is the difference in rating. Since a greater difference in rating would indicate a more divisive movie, this could explain the negative correlation. Again, this is very slight.

Fig. 6.2: *Primary Actor (with at least three films) vs. Box Office Performance*



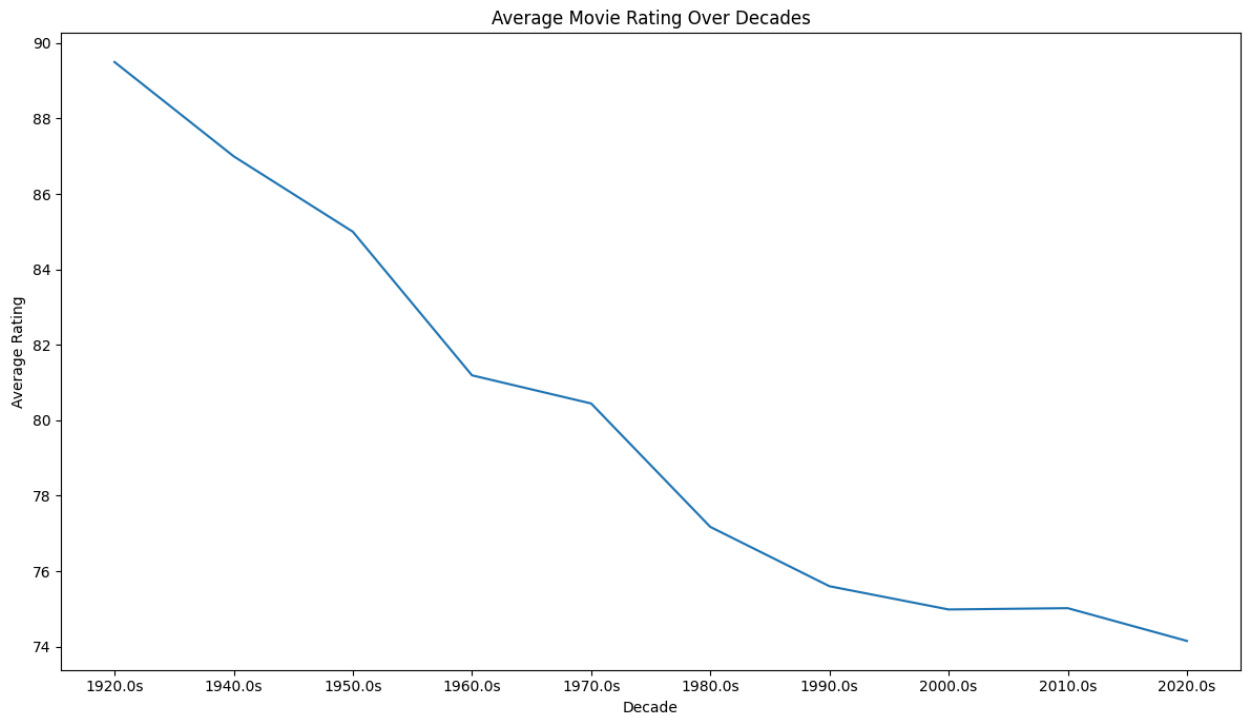
This compares the primary actors, who have been in at least three films, against the average box office performance of their films. Based on this chart and the input data, Christian Bale would appear to be the highest-paid actor. This is likely because our data set contains the Dark Knight trilogy, which fared quite well. Looking down the list, we see many of the usual names that we would expect to fare well at the box office, such as Kevin Spacey, Robert De Niro, Al Pacino, and Gene Hackman. Based on this performance, producers may want to consider casting some of these A-listers to potentially capture higher revenue at the box office.

Fig. 6.3: *Director (with at least 5 films) vs. Average Box Office Performance*



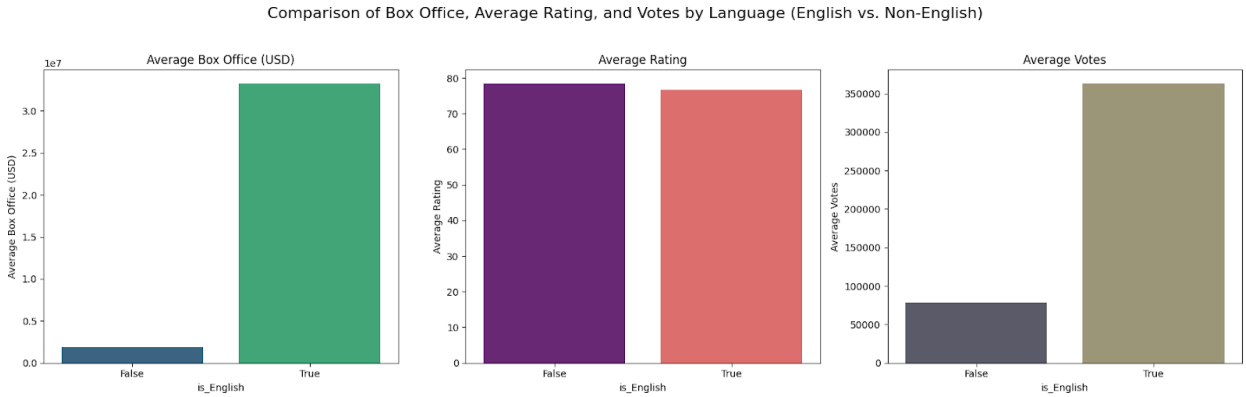
This chart displays the average box office performance of various directors who have directed at least five films on the list. In a similar vein to the last graph, Christopher Nolan, the director of the Dark Knight trilogy, blows the other directors out of the park with box office performance. Surprisingly, big names like David Lynch, Park Chan Wook, and Krzysztof Kieslowski captured relatively low revenue at the box office on average, but their films tend to be more niche and less appealing to broader audiences.

Fig. 6.4: *Average Movie Rating by Decade*



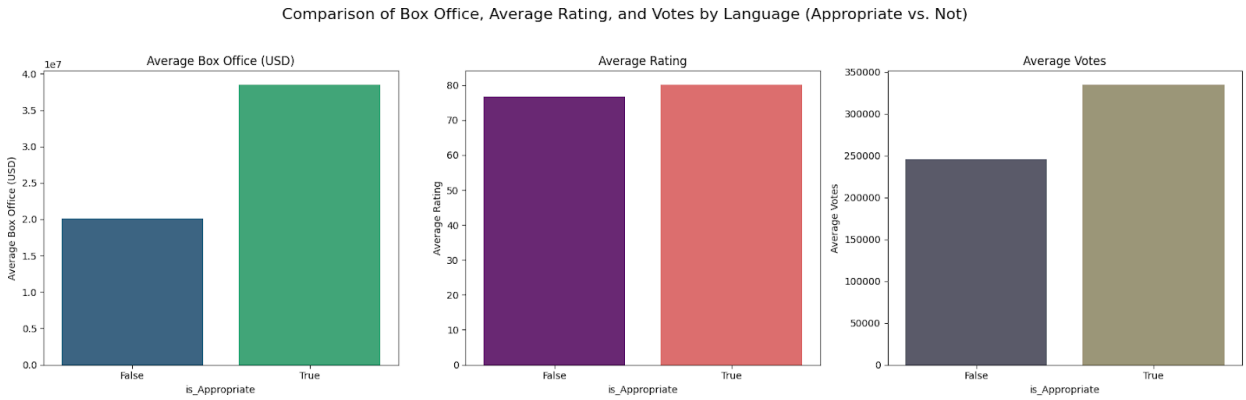
This chart displays the average rating of films grouped by decade over the past 100 years. One key consideration here, though, is that survivorship bias may play a heavy role in manipulating this data, as our list likely only contains the objectively good films from the early 1900s, while having a more diverse spread of ratings for more recent films. Past that, this consistent downward trend could simply be explained by nostalgia for older films, and some bias in the viewers that sees these films as objectively better due to their age or cultural significance. Regardless, it was interesting to see this consistent decline in ratings over the years.

Fig. 6.5: *Average Box Office, Rating, and Votes by Presence of English*



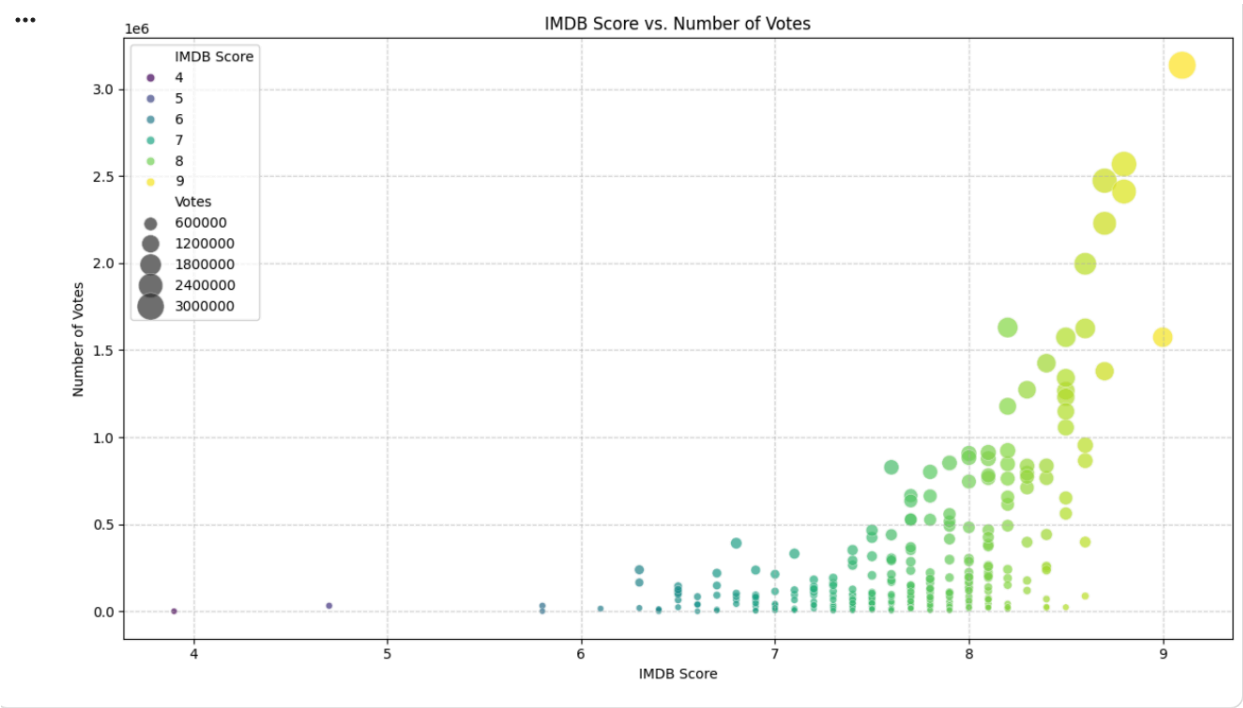
These three figures compare average ratings, votes, and box office performance by whether or not films were offered in English. Starting off with box office performance, we see a huge difference, though this is likely explained by the fact that this statistic is only for US box office performance, so English films would typically fare better. Moving on to average rating, the two categories performed similarly, though the foreign films actually tended to perform slightly better, which could again be due to some level of survivorship bias in the creation of the dataset. Finally, looking at average votes, the films offered in English tended to be more popular, which again makes sense as most users of IMDB are probably American, or at least prefer to watch films offered in English.

Fig. 6.6: *Average Box Office, Rating, and Votes by Age Rating*



These three figures compare average ratings, votes, and box office performance by whether or not films were deemed categorically appropriate for children. Unsurprisingly, films that were rated more appropriately performed better at the box office than those rated R or worse, as they likely drew in a larger crowd. Interestingly, though, from the perspective of someone more interested in film, the age-appropriate films actually performed better on average as well when it came to ratings and average votes. This surprised us as usually there is this idea that the objectively better films and those that tend to be more popular, at least with an older audience, are those that are often more intense and rarely meant for children. Seeing this trend was therefore quite surprising and made an interesting discovery.

Fig. 6.7: *Do Higher Rated Movies Attract More Votes on IMDb?*



This compares IMDb score with the number of votes, taking into account the coloring, which aids in the visual representation. One can see the direct increase in rating as the number of

votes increases, which is not all that surprising. Better movies obtain better IMDb scores, and this causes more people, including critics, to watch them and rate them. There are no obvious outliers to this expected relationship between variables, but we were still surprised

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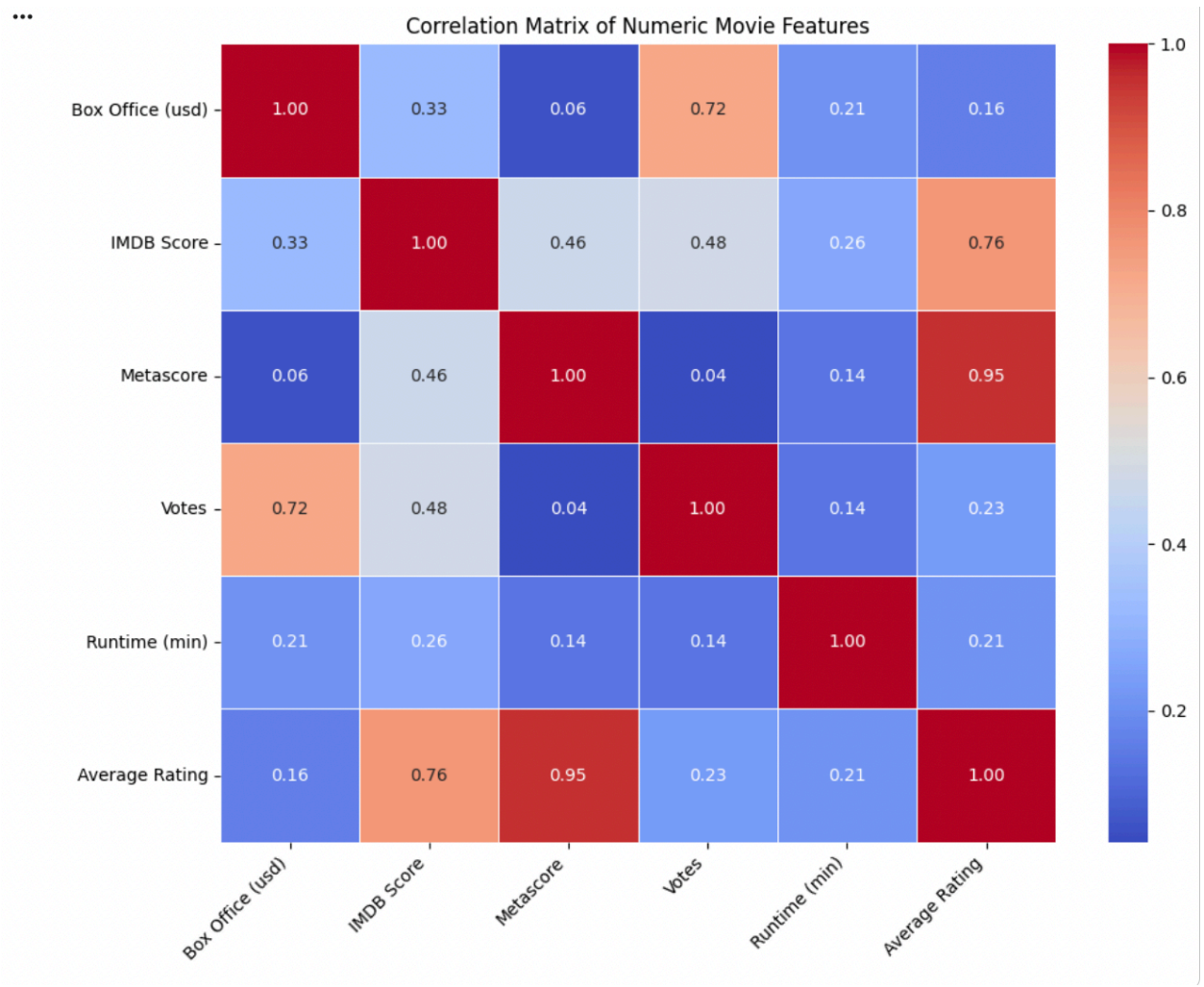
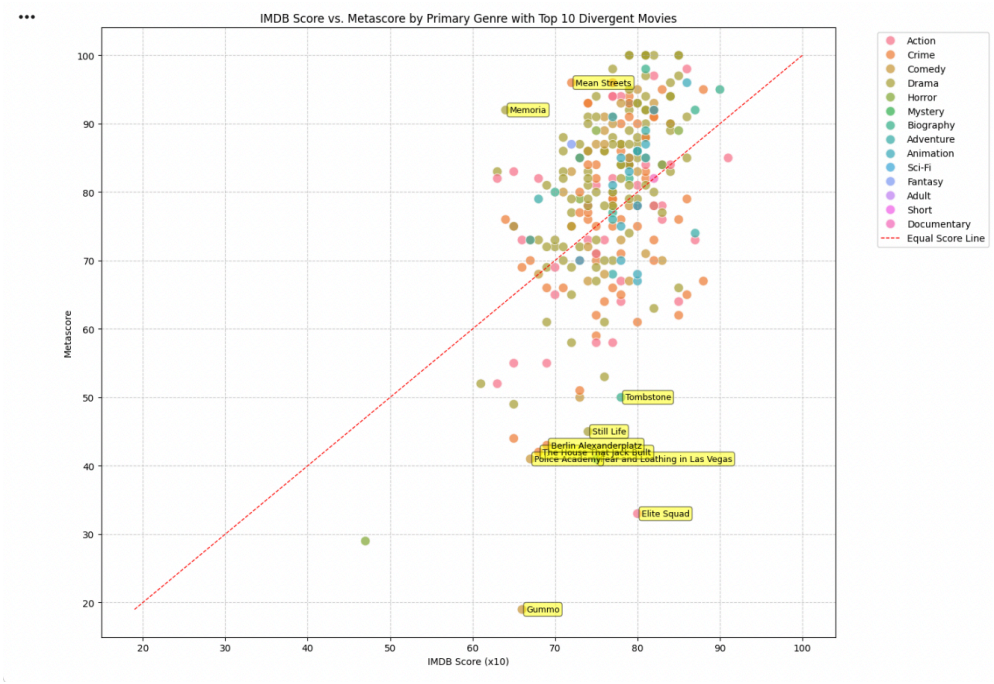
Fig. 6.8: *Correlation Matrix*



This compares each variable that is being tested during this process, showing its relative relationship with each other, colored blue and red for help reading. With the average rating being the weighted combination of Metascore and IMDB Score, it makes sense that there is a stronger

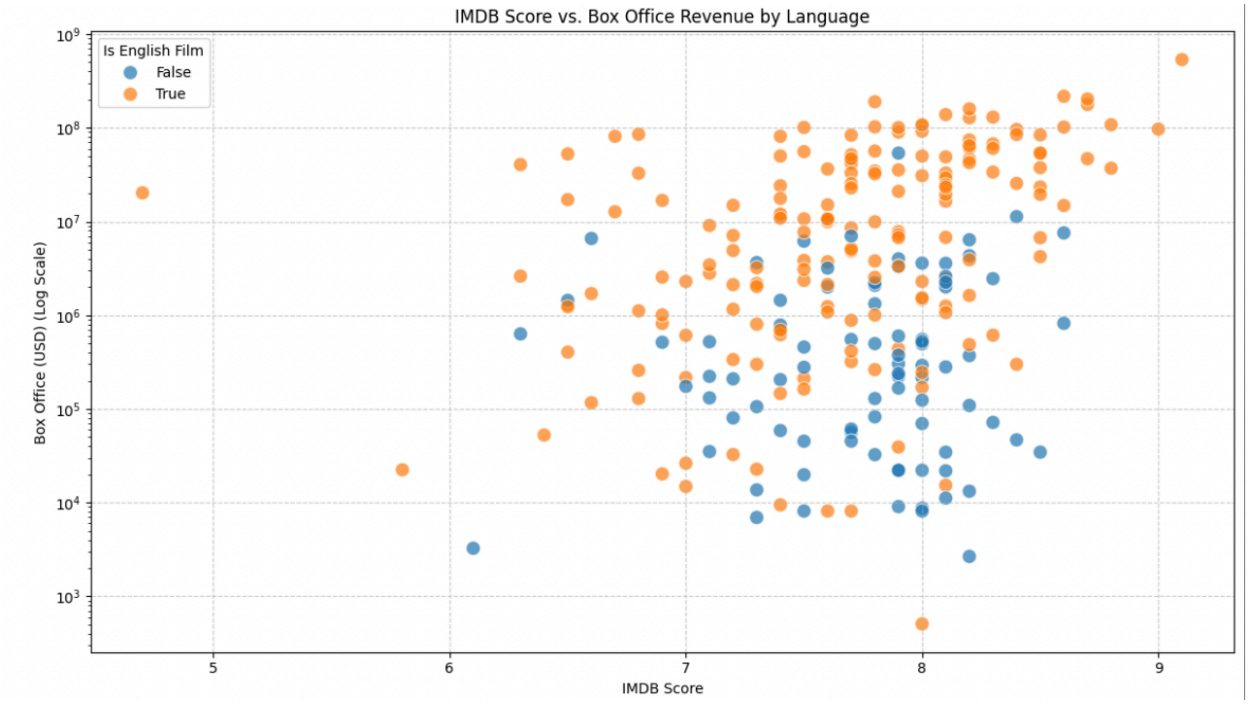
correlation between these variables. Another relationship that demonstrates a strong correlation is box office and votes, which is graphed in 6.1. This graph also shows a surprisingly low correlation between IMDb Score and Metascore, which shows the humanity in the subjectivity of movie ratings, and how sometimes it's better to just decide for yourself.

Fig. 6.9: Do Critics and Audiences Agree on Film Quality, and Which Films Divide Them Most?



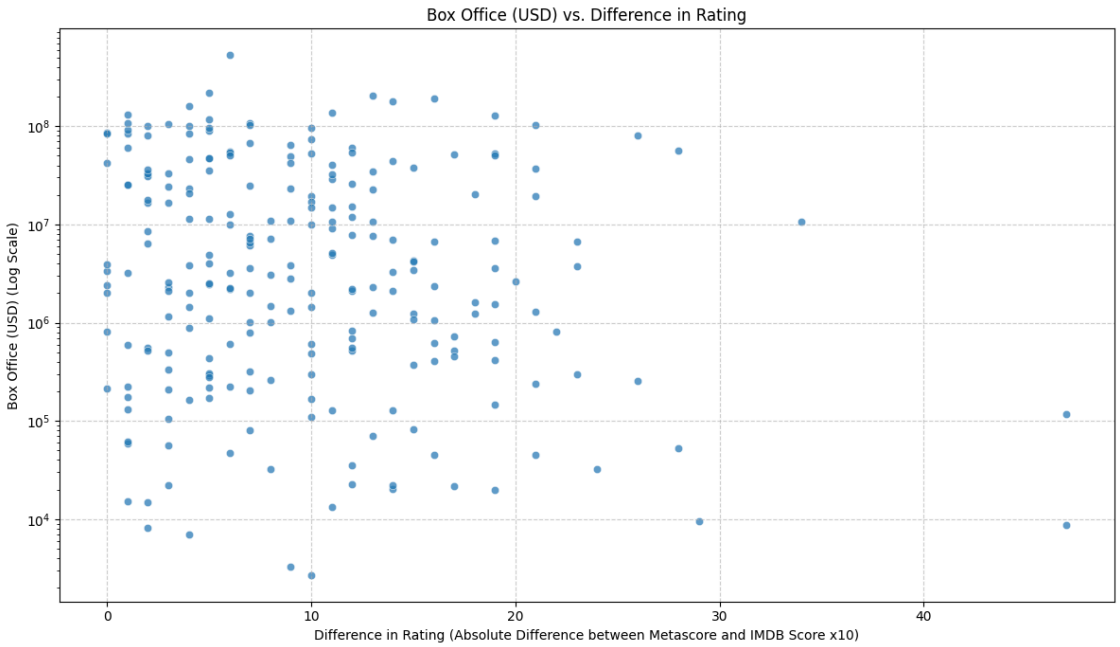
This graph compares Metascore and IMDb Score by genre, pulling out and naming the biggest outliers. There is an obvious cluster between 75-85 in IMDb Score and 65-90 on the Metascore. As IMDb is an audience-backed rating system, this suggests that critics vary in their opinions more than fans. Gummo is the most obvious with the biggest distinction between the variables, while Tombstone was the first one I noticed, known for its quality, to be so relatively low in Metascore.

Fig. 6.10: *Does Being an English-Language Film Affect Box Office Performance, Independent of How Well-Rated it is?*



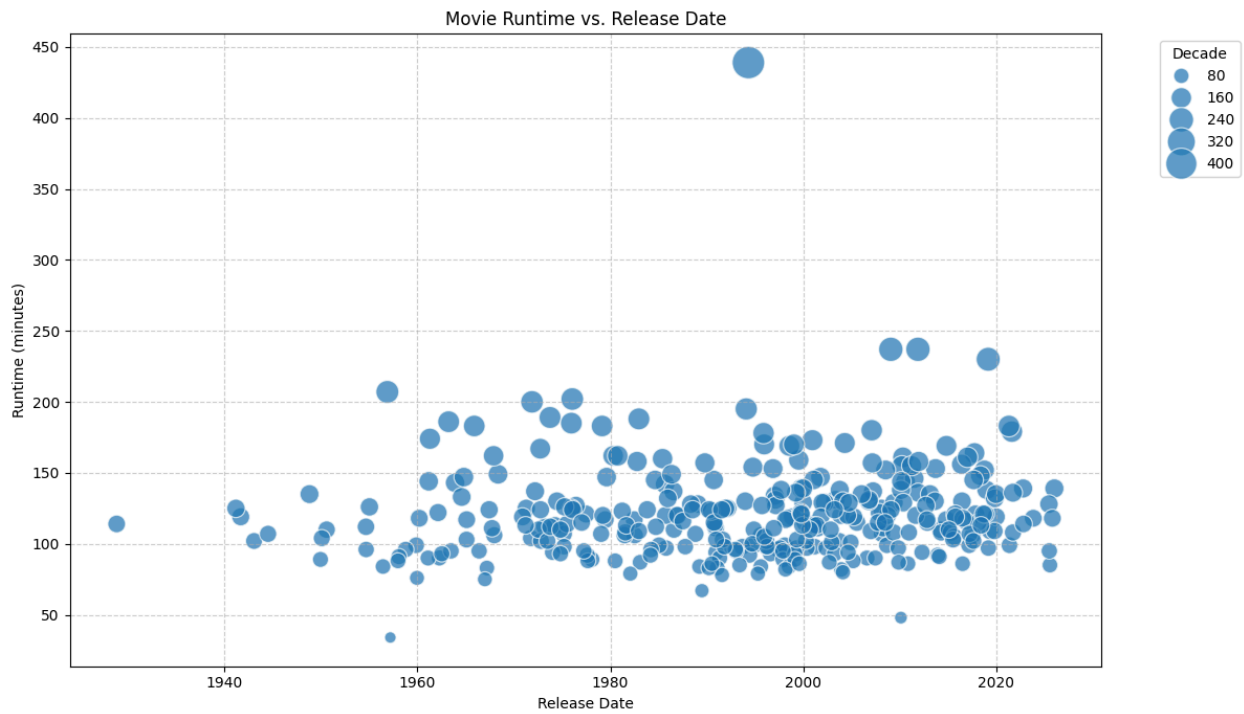
This graph shows IMDb scores against the box office, with the blue representing non-English movies and the orange representing USA movies. Very obviously and predictably, USA-made movies performed better at the box office than non-English films. The one obvious outlier in this graph is the non-English film at a 4.5 IMDb score, but with a relatively high box office performance.

Fig. 6.11: Box Office (USD) vs. Difference in Rating



This chart is a scatterplot displaying the absolute value of the difference in rating for the films against the log of their box office performance to normalize for outliers. Based on this chart, there is not a terribly strong correlation between the two variables, though it appears that the two clear high outliers in difference in rating fared more poorly than most of the films on the list at the box office. This graph suggests that whether or not critics agree on IMDB and Metascore regarding their ratings for the film is somewhat independent of how well that movie performed at the box office.

Fig. 6.12: Movie Runtime vs. Release Date



This figure compares the release date of a film against its runtime on a scatter plot. Looking at the trend in the data, there does not really appear to be a relationship between runtime and the release date. Perhaps there is a very slight tendency that more recent films are longer on average, but the difference is relatively insignificant. This shows, somewhat surprisingly, that movies have not really changed too much in runtime from the earlier days, or at least our selection of them.

7. Discussion

With this project, there were a few important limitations to consider. For one, the box office variable only reflected how films performed in the United States. This was an obvious weakness for foreign films, especially the older ones, as some of them did not even have data for box office performance in the United States. Ideally, we would have data on global box office performance, not just the US information alone. This would have given a more accurate view on the relationship between different aspects of a film and its box office performance. Additionally, the dataset itself

was relatively small at 338 films and was hand-selected, which gives room for ample bias. Our reasoning for this data set, though, despite its limitations, was a 1000 calls per day limit on the free version of the API. For future projects, it would be ideal to be able to make more daily calls to the API with a more random data set. We actually found a very large list of films, which included over 61,000 entries. However, given the power of our systems and the number of data points, this was not realistic to use.

8. Conclusion

The exploration of 338 movies through the OMDB API resulted in a pretty consistent finding that votes (popularity) predict box office better than critics' scores, suggesting that great movies sometimes miss out on popular appeal, but that movies that obtain more votes/have closer attention, predict box office pretty well.

One of the more interesting findings we found was that non-English films with the same or higher ratings were very far off when compared to the box office returns from English counterparts. We also found that critics' scores varied more than the audience's scores, which works as a critique of the subjectivity of critics' ratings.

It's important to note that this list was handpicked, so there can be an acknowledged sense of skepticism when it comes to concrete answers to these questions. The non-English movies showed that although their performance at the box office was relatively low, their quality stood the test of time and difficulty, making its way to us.

Overall, this project gave us ample insight into how the characteristics of different films relate to their relative popularity, success, and perceived quality. While some findings were shocking, others proved our assumptions, and a few showed little correlation at all, we surely learned a lot about film performance through this EDA exploration.